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A2

# Dataset: World Bank Data Indicators

This dataset contains statistics for several key data indicators throughout the world, broken down by country and year. It provides key fields for climate change, energy consumption, trade, poverty, and many more areas. Since there are 20+ unique indicators, I've decided to focus on a few of them to make my analysis more robust and focused. These indicators are: climate change, energy and mining, and economy and growth. I am interested in the complex (or perhaps not so complex) relationships between these key indicators (see my Initial Analysis Questions below).

This data was originally provided by the World Bank here. The data I use to generate the visualizations below was taken from a post-processed set of data files, which can be found in this GitHub repository.

# Initial Analysis Questions

1. **Overarching Question:** What are key correlated features between climate change, energy usage, and economic growth?
2. What is the direct or indirect impact of certain energy and mining practices on a country's finances?
3. Are there clear connections between increased energy production and visible climate change effects?
4. Are there countries with high economic growth that do not have a significant effect on climate change?

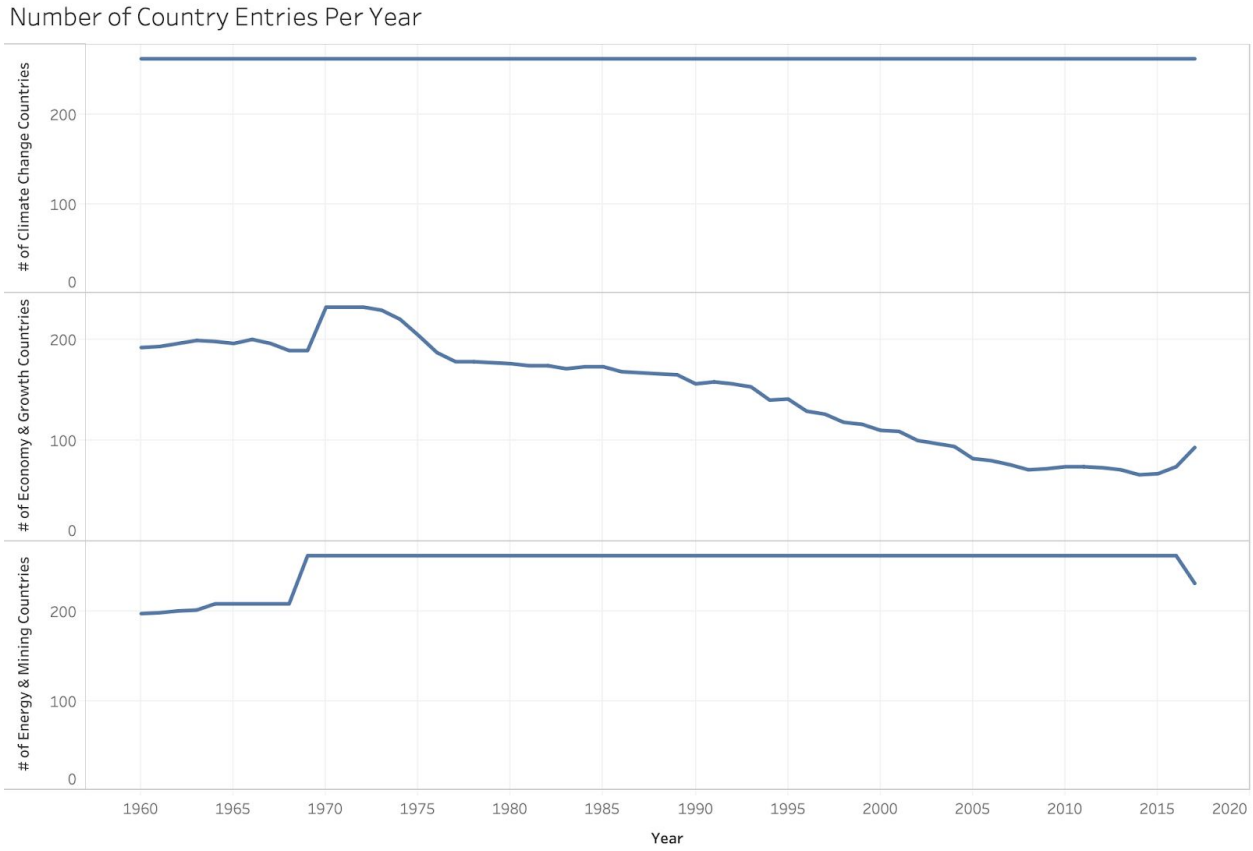
These questions were posed before even digging into the dataset with any real specificity. More questions arise throughout my work, which are documented below.

# Discoveries & Insights

First, I import the three relevant datasets into Tableau and join them via the **Country Code** field, which is a unique and shorter identifier for each country in the world that the World Bank uses throughout many of their files.

Each data file also has several entries for each country, with each entry corresponding to a particular *year*. I also join the data based on this **Year** field, so the different data sources are also aligned in terms of when these data points were gathered. First, I felt it was important to get a sense of the distribution of what years each of these 3 tables has data for. In particular, I

simply plot the year on the x axis vs the number of rows with that given year in each table on the y axis.




The trends of count of climate-change.csv, count of economy-and-growth.csv and count of energy-and-mining.csv for Year.

It is immediately clear that while the climate change dataset is extremely dense and has a full set of 263 countries for every year from 1960 to 2017, the other two datasets are not quite as full. The energy and mining dataset has 263 entries for a majority of years (1969-2016). The economy and growth dataset, on the other hand, only has at best 232 country entries at its peak in 1970, falling off from there. This information is critical as I cannot just blindly assume I have data for every country, for every year.

This plot also indicated that we may have some null values for the Year field, so we always filter out those for subsequent plots in the analysis.

Since we are missing so much economy and growth data for certain countries, I have to make the tough choice of how to handle this. There are a minimum of 66 countries in a given year in the economic dataset, based on the above plot. Next, I plot the number of years we have economic data for each country (sort of the inverse of the plot above), using a size encoding on each geographic region. Surprisingly, we have data for each country for at least 4 years, and some up to 58 years. Therefore, it is likely that *different countries have data points at different*

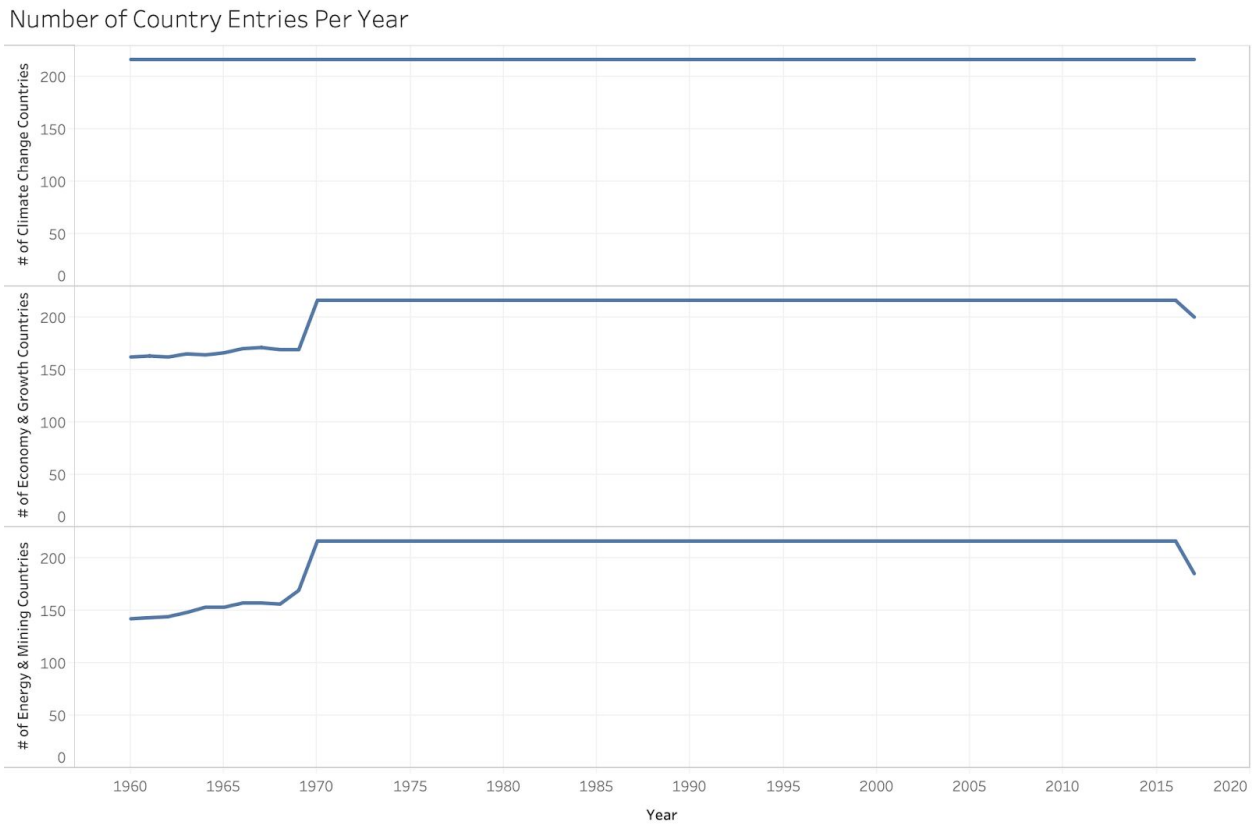
*frequencies and offsets*. This is reassuring, because otherwise we would have to discard countries altogether.

Number of Year Entries Per Country, Economic Data

Map based on Longitude (generated) and Latitude (generated). Size shows count of economy-and-growth.csv. Details are shown for Country Code (economy-and-growth.csv).

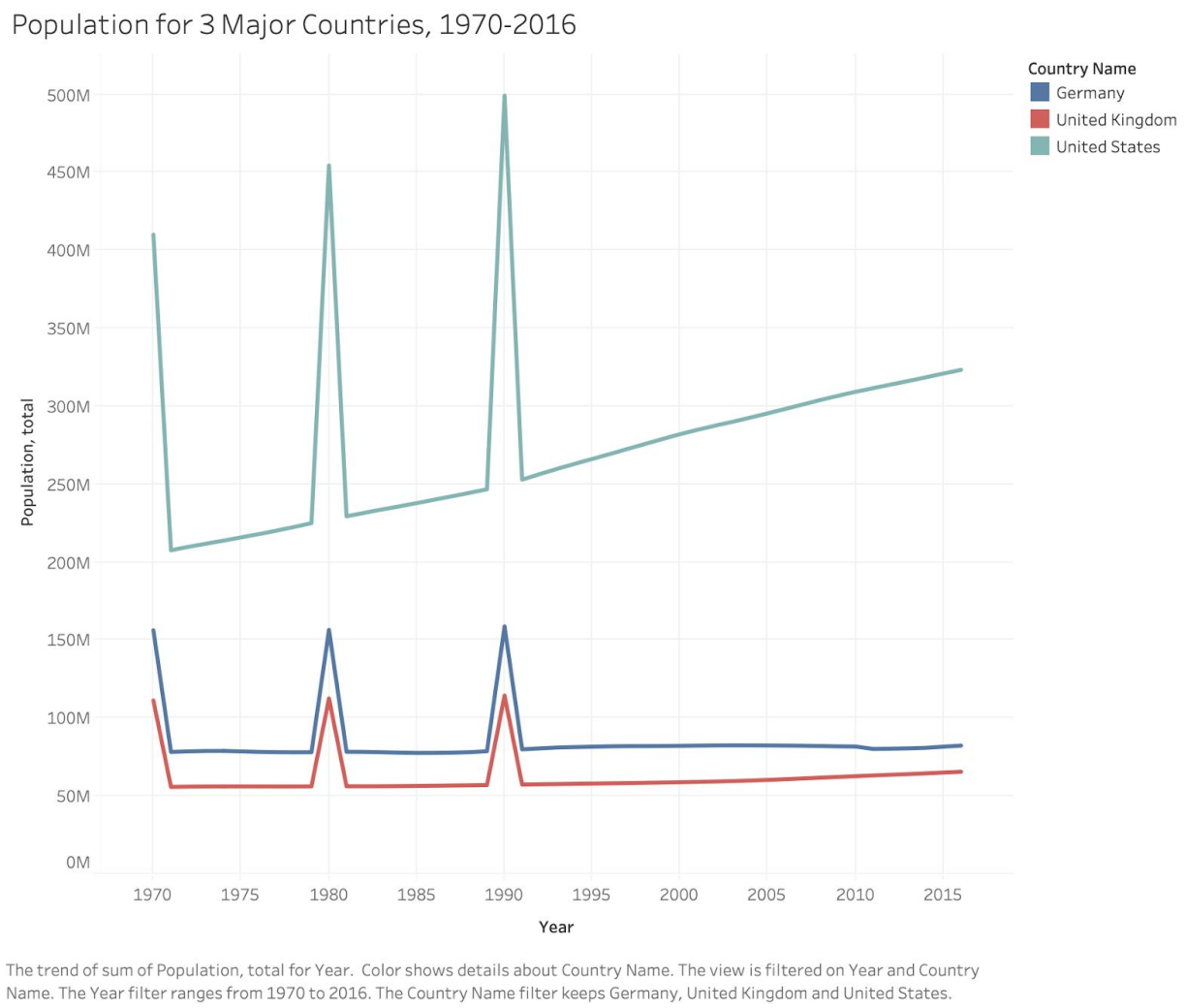
Upon generating this visualization, we also get a warning that 47 values in the dataset have unknown geographic locations. So, moving forward, we always filter out these values to avoid having these discrepancies. Upon visual analysis, many of these values are not even countries, but rather summary descriptions of regions that have been created by the World Bank dataset owners. Some examples include “Caribbean small states,” “European Union,” “IDA total,” and “North America,” to name a few. For all future visualizations, we filter out these invalid country names and codes (I did this on my personal machine with some quick Python scripting and generated new CSV files, which you can find in my fork of the GitHub repository here. I did try to do this in Tableau first as you advised here, but given that there were 47 manual values to filter, I went for a faster automated approach!) This is just one great example of how **visualizing your data can allow you to quickly find inconsistencies and errors that require additional processing.**

Interesting! Now, when I regenerate the first plot on the filtered dataset, it seems like we have data for all countries of interest from 1970 through 2017, with the tail ends of the year distribution missing some data.

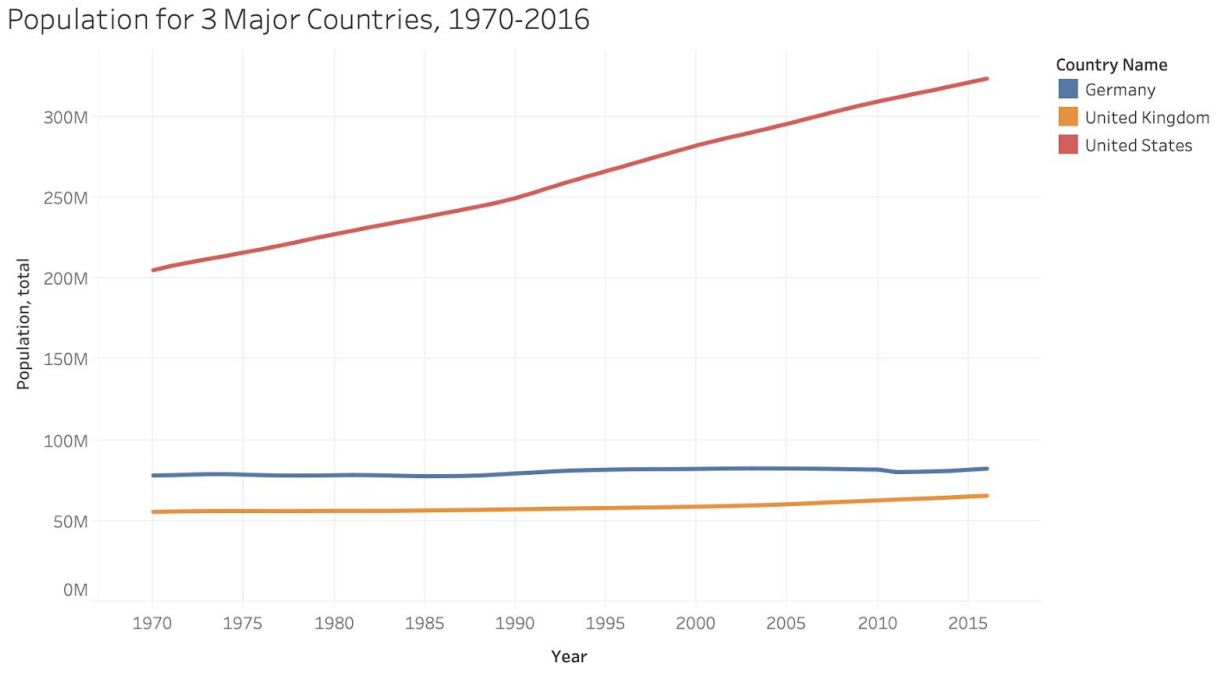



So, for all future visualizations, I also add a filter on **Year** so that we only consider 1970 through 2016. This still gives us a pretty complete picture of the dataset, while making things simpler so we can always compare our 3 indicator categories for every single year. (For instance, if I do not use this filter, then some countries will not have data for say the year 1965, which could skew our analysis slightly.)

Great! Now that we have made some key preprocessing steps, let's dive into some other basic facts about our data. First, for example, let's take a look at the population of 3 major countries from 1970 to 2016.




Woah! Those spikes in the years 1970, 1980, and 1990 sure look strange! Turns out this is actually a well documented issue in this dataset, as confirmed on Slack and a GitHub issue on the original repository where we sourced the data. Again, I do some quick preprocessing in Python to get this right:



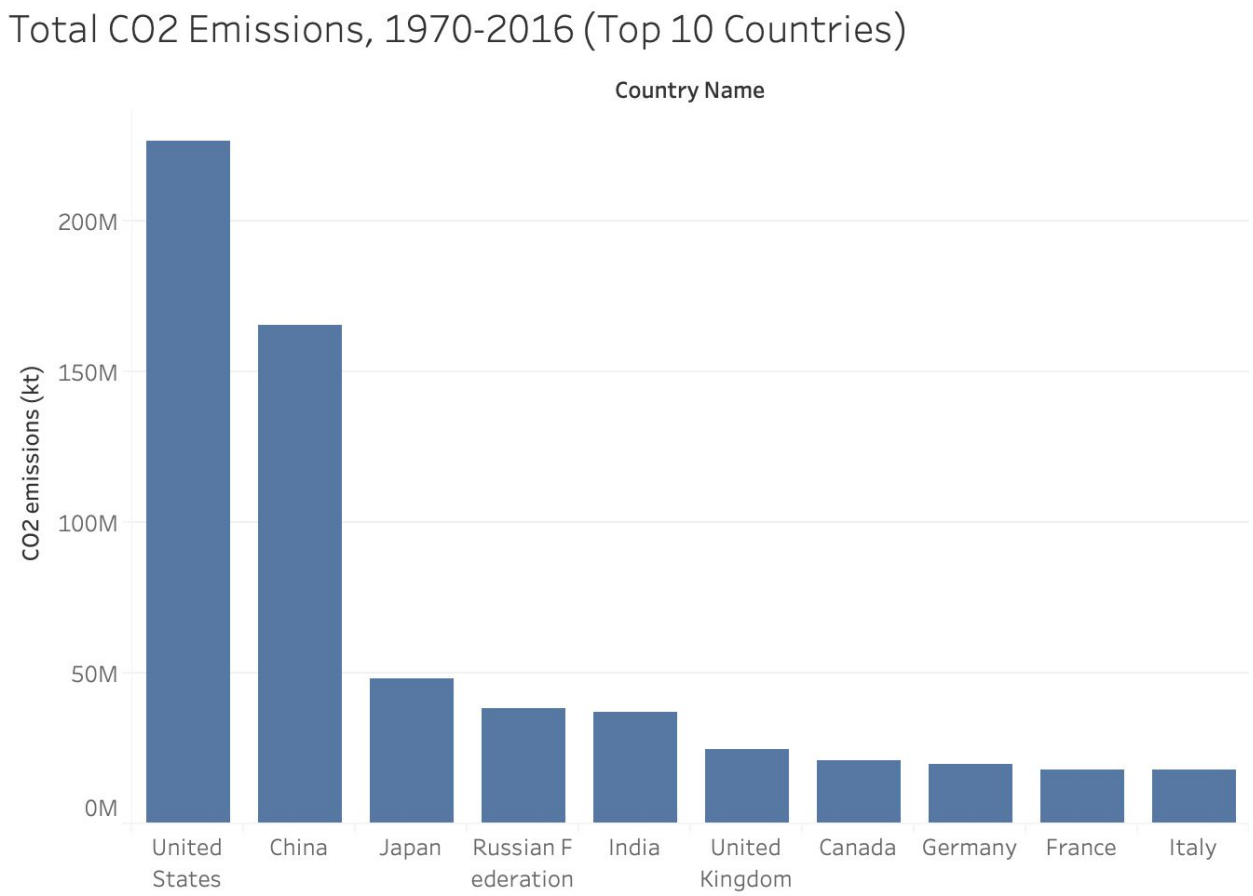

The trend of sum of Population, total for Year. Color shows details about Country Name. The view is filtered on Year and Country Name. The Year filter ranges from 1970 to 2016. The Country Name filter keeps Germany, United Kingdom and United States.

Great. Before diving into more complex relationships between climate, energy, and economics, I wanted to look at some other univariate distributions of interest. Here, I note the rise of CO2 emissions per capita in the non-Western world as we progress from 1970 to 2014. This trend is an important one for me to recognize early as it demonstrates the global path climate change has run in the past 40+ years.

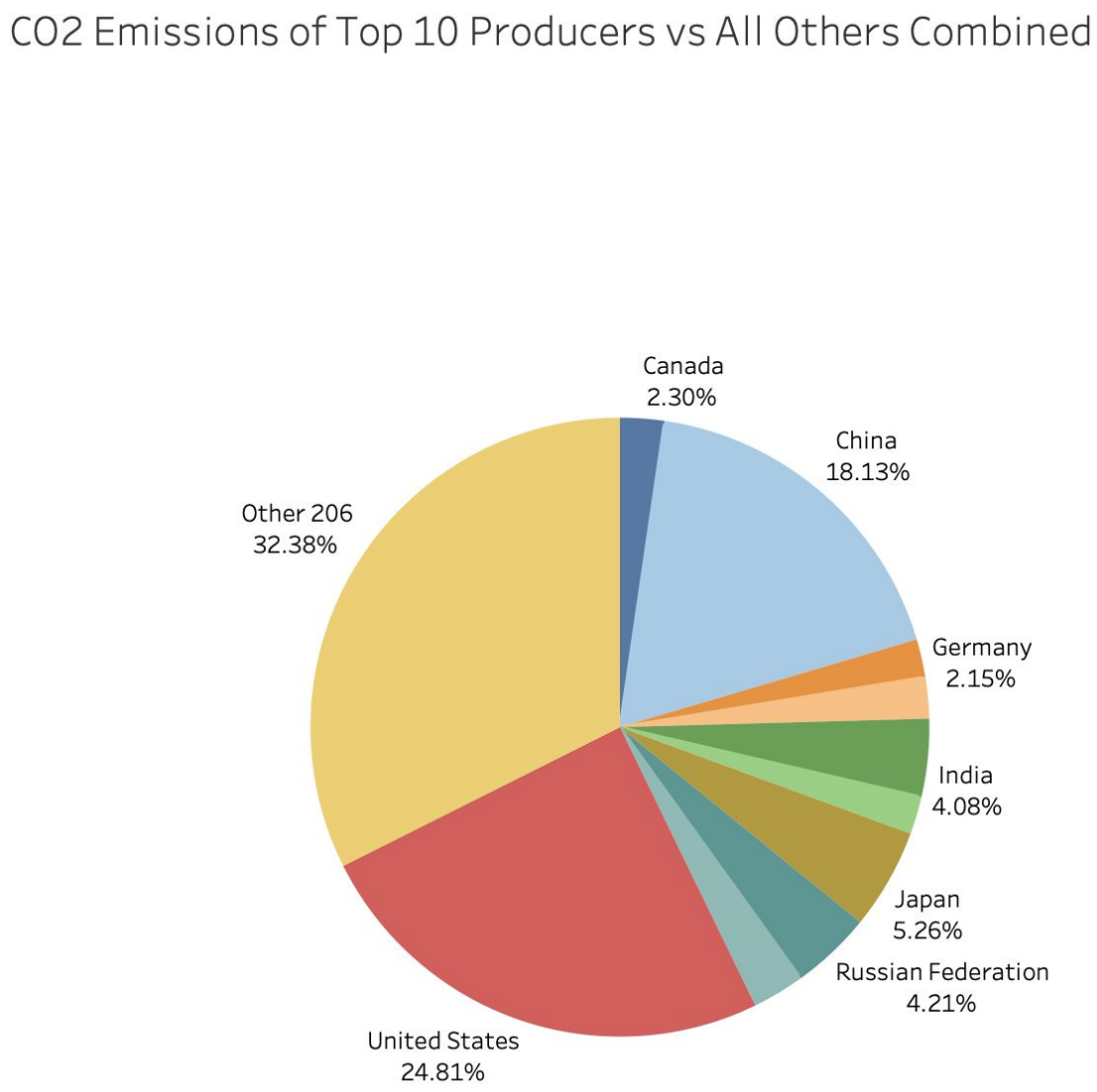
CO2 Emissions Per Capita, 1970 vs 2014

Map based on Longitude (generated) and Latitude (generated) broken down by Year. Color shows sum of CO2 emissions (metric tons per capita). Details are shown for Country Name. The view is filtered on Year and sum of CO2 emissions (metric tons per capita). The Year filter keeps 1970 and 2014. The sum of CO2 emissions (metric tons per capita) filter ranges from 0.00 to 70.00.

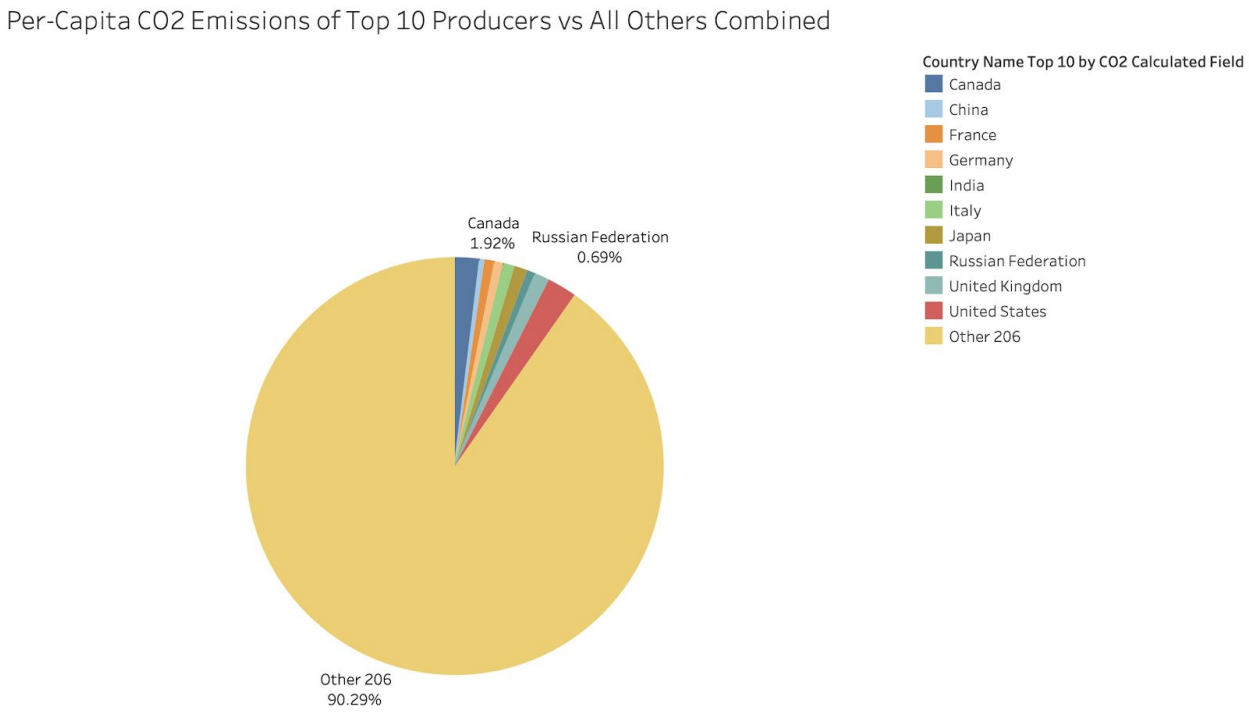
A natural next question to ask is the following: which countries are contributing the most to these CO2 emissions? A simple bar chart with the top 10 CO2 producing countries provides a quick answer:



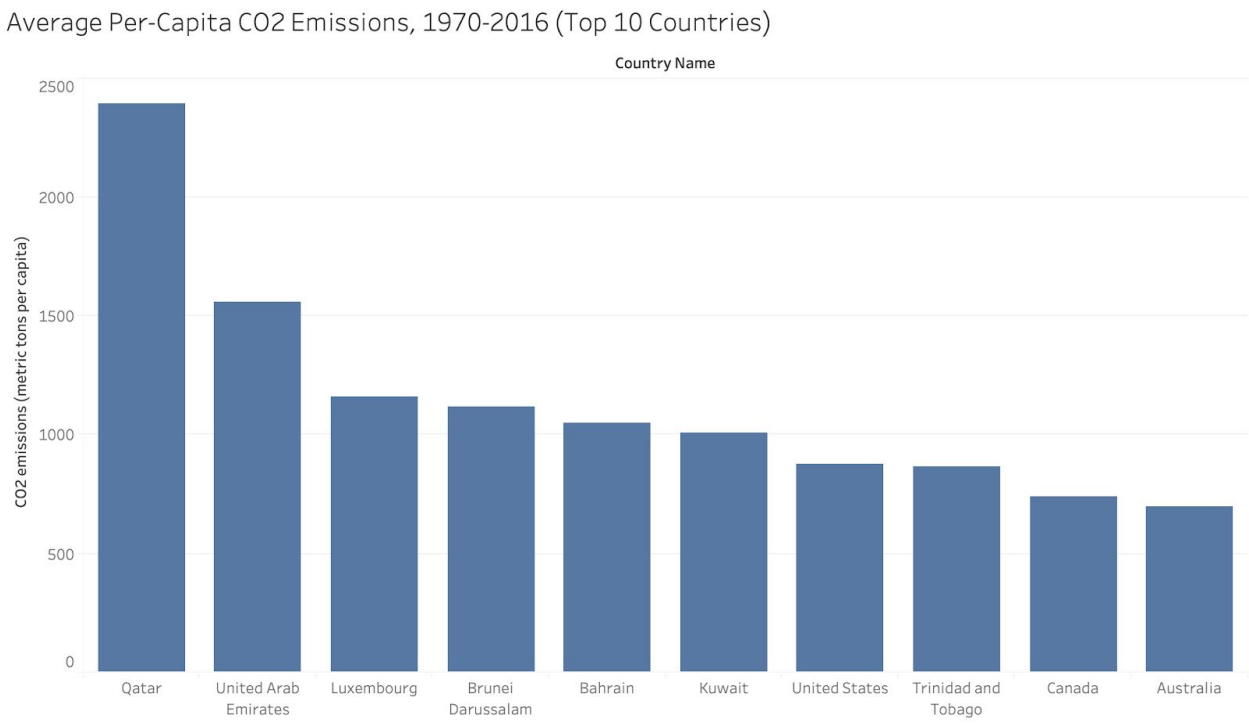

And we know that many things in our modern society are very **non-linear**. In this case, it would be great to know how much of the total CO2 emissions in the world are generated by these top 10 countries. For this purpose, I use a simple pie chart to compare the top 10 countries against all others combined in terms of CO2 output in kt (kilotons), in the year 2014 only for simplicity (the latest year for which we have data on this particular field):




Wow! So, it is immediately clear that the top 10 producers of CO2 emissions produce over ⅔ of the total emissions worldwide! This imbalance is striking. And what is even more compelling is how much of this makeup is dominated by the United States and China. What is worth noting, though, is that this value is not normalized per capita. When we take into account population, here is the new distribution:

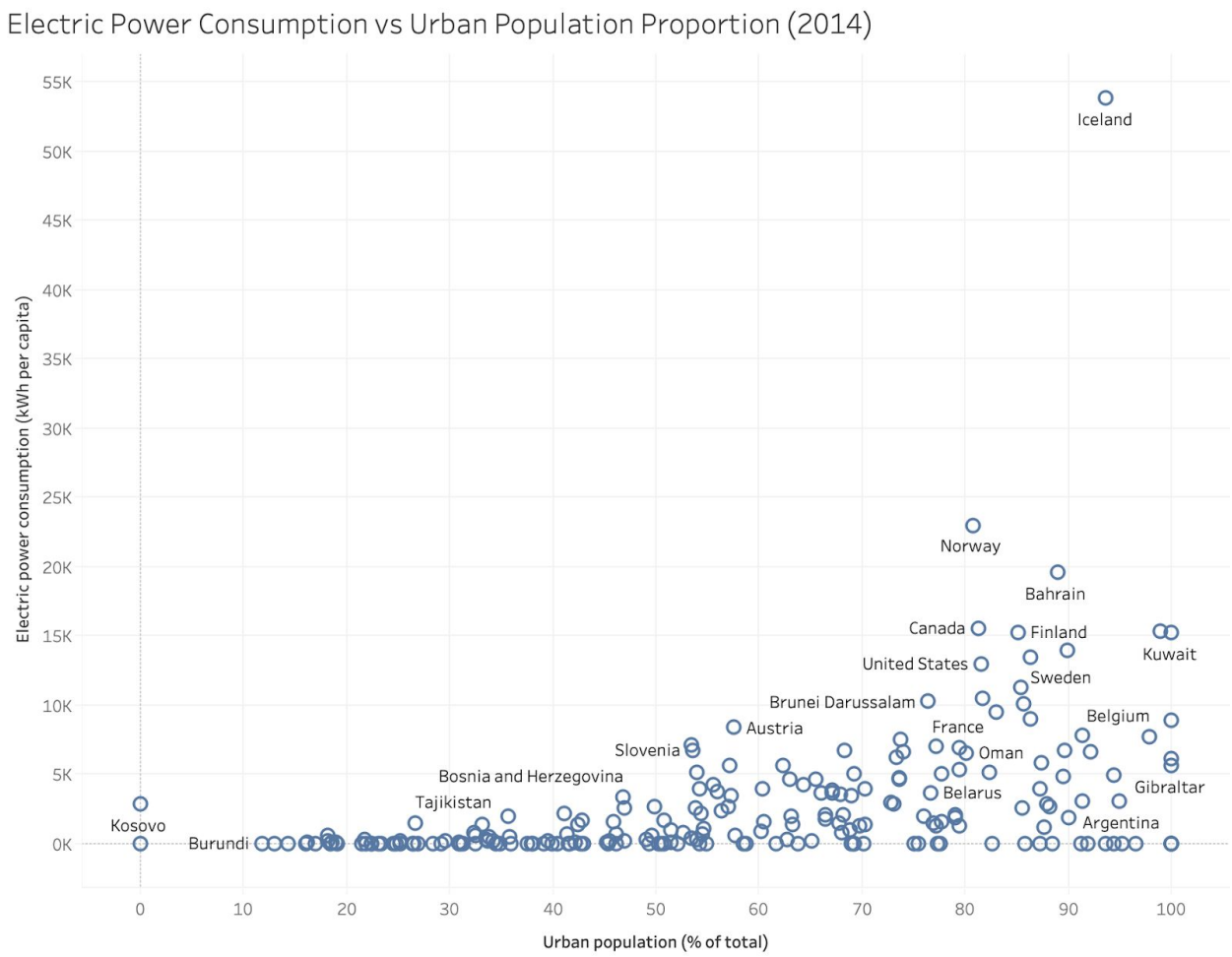



This is quite striking. On a **per-capita level**, the top 10 producers are actually not contributing that much more than its relative expected share. Who is doing poorly on a per-capita level, then? While you cannot simply normalize for population as the only measure of how a country is doing with emissions, it can certainly be eye-opening. Let's recreate a simple sorted bar chart of CO2 emissions as we did before, but on a per-capita level now:



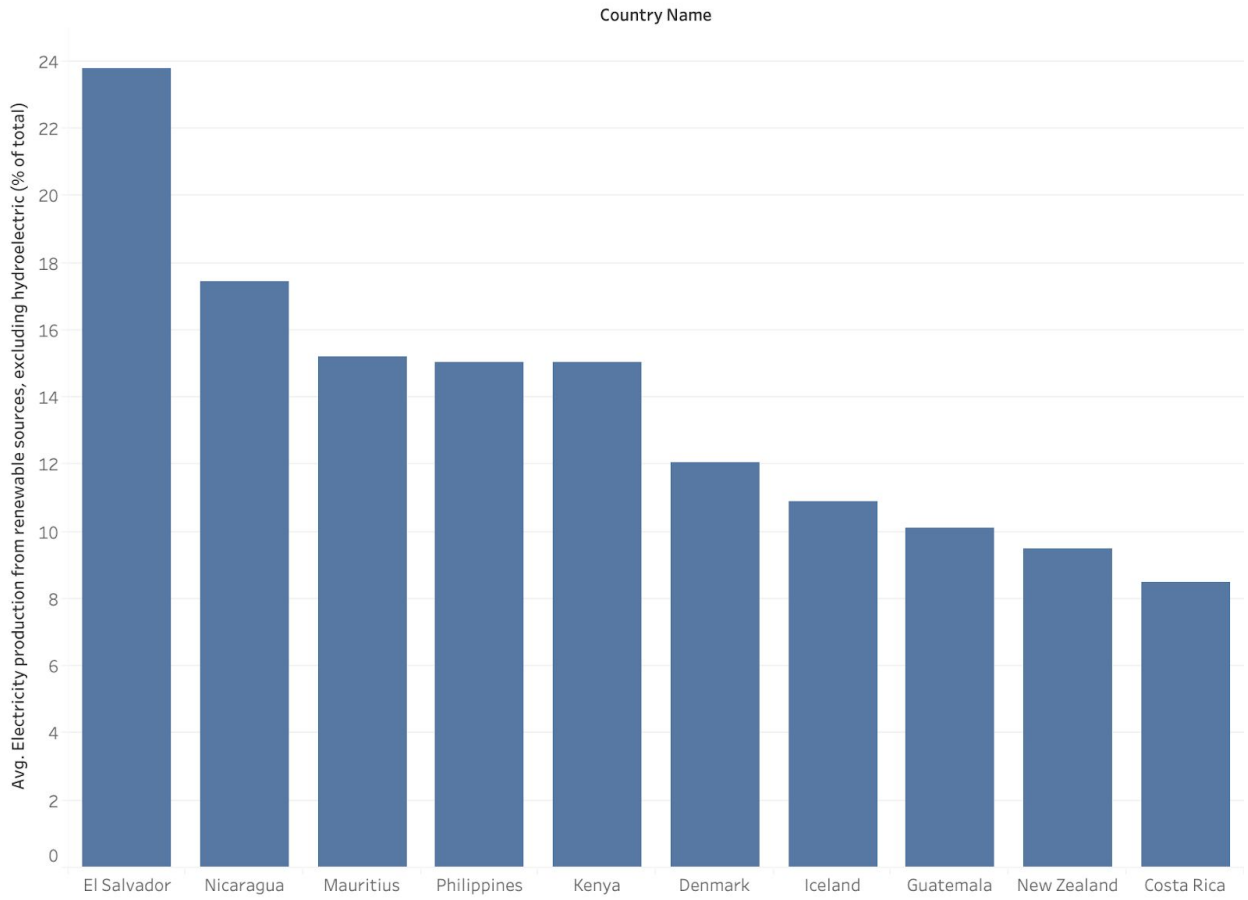

This makes a bit more sense - there are 6 countries that fare worse in CO2 emissions than the United States on a per capita level, making them interesting areas for further analysis. But, we still see the United States as the 7th worst since 1970 in this area - clearly we have lots of room for improvement.

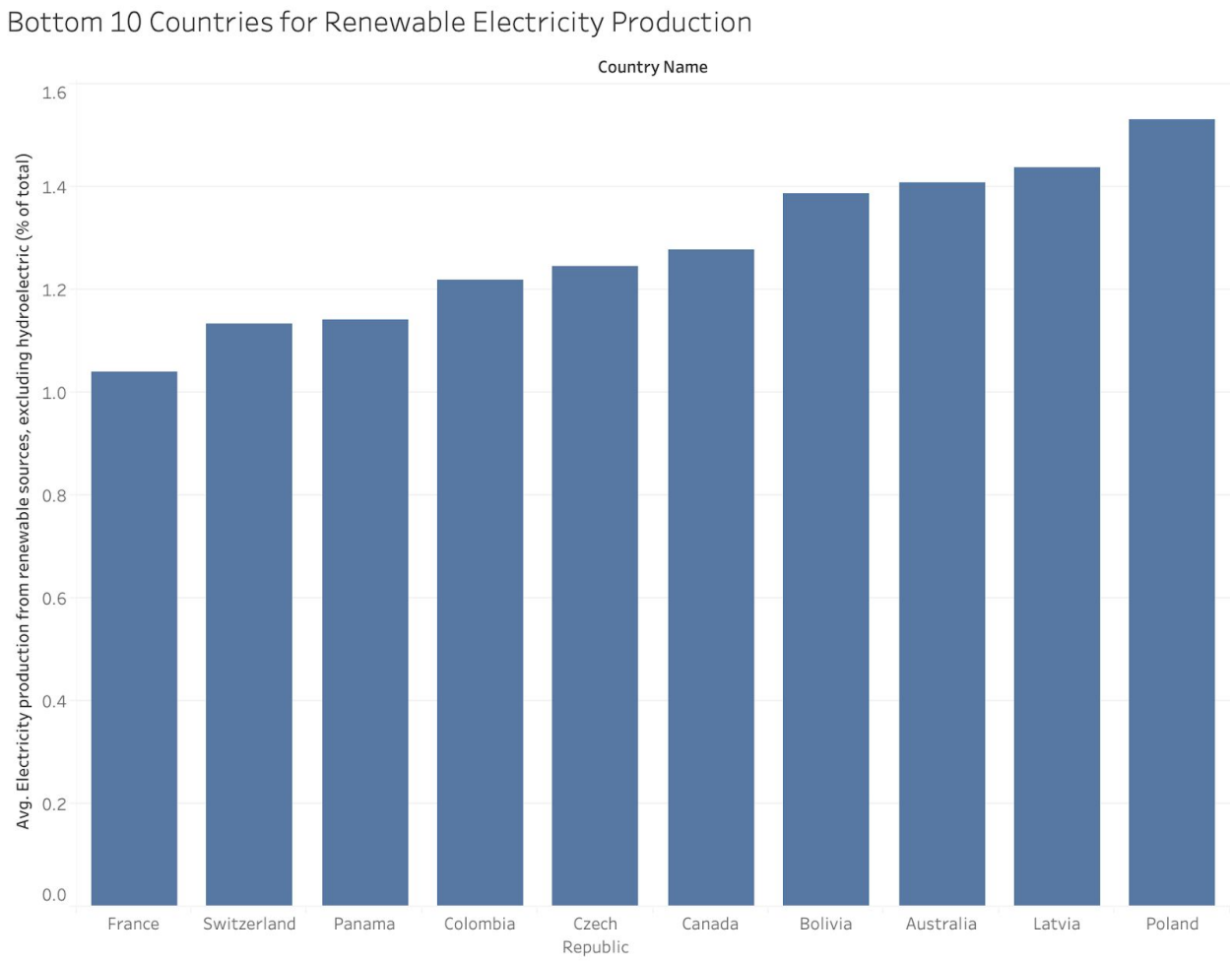
Now, let us integrate some key information from the economic growth and energy/mining datasets also provided by the World Bank. Below, I combine the climate change dataset's measure of the proportion of a given country's population living in an urban area to the per capita electric power consumption. (Notice how I learned my lesson before about not taking per-capita effects into account.) There are a few key things to note here. First, there is a clear positive relationship where more urban countries have citizens who consume more electric power. What also stands out though is the striking outlier of Iceland in this chart - nearly all of their population is in an urban area, but their electric power consumption per capita is more than double that of the next closest, Norway.




This striking difference for Iceland as an outlier makes me wonder - just *how* are these countries producing their energy? In particular, can we say something about *renewable* energy production and how that may differ between the “best” and the “worst” countries? The two plots below provide some early insights.

Top 10 Countries for Renewable Electricity Production

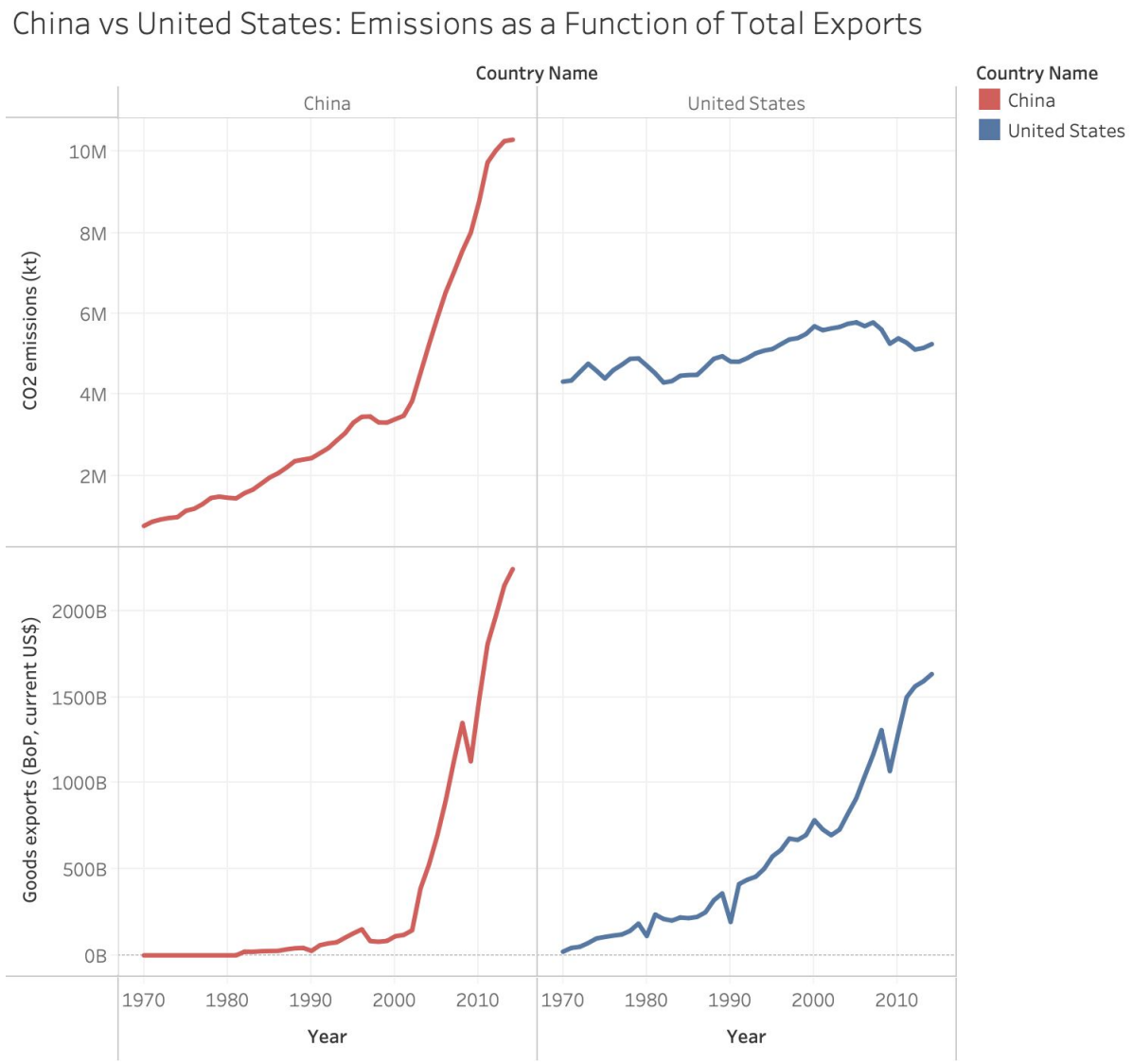





There is a striking difference between the top and the bottom here. As you can see, the top 10 countries have anywhere from 10 up to 20+ % of their total energy production come from renewable sources. Compare that to the bottom 10, where they have only 1 or 2 % coming from renewable. What is even more striking is that **165 of the 216 total countries had an average of ZERO percent of their electricity produced by renewable sources, all the way from 19700 to present day.** The visualization directly above this represents countries 166-175 when sorting that renewable percentage ascending. This just goes to show the significant number of countries who have **no renewable** electricity production whatsoever, and just how much work needs to be done in this space.

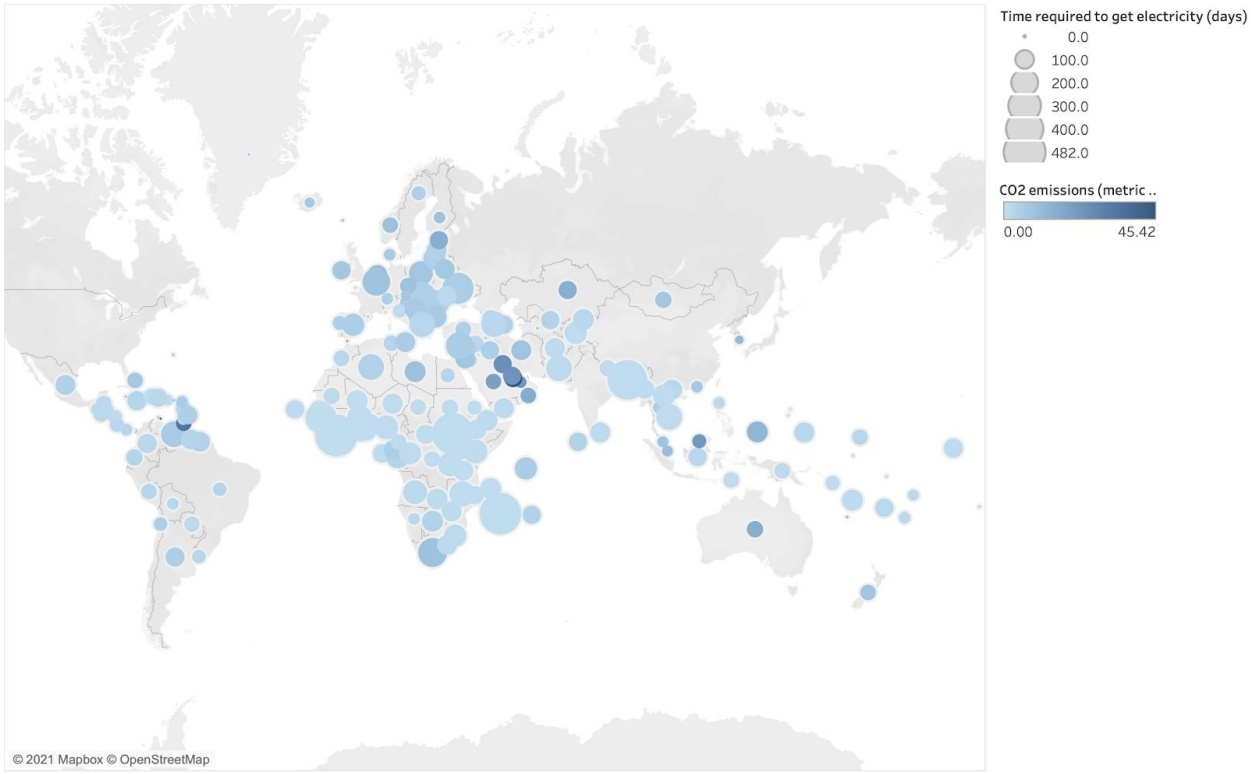
One can only ask - how can this be? How can some countries be so lacking in renewables? What drives other countries to have the energy infrastructure they have? One way we can look at this is in terms of net exports for a given country. The figure below highlights the relationship between total exports and total CO2 emissions over time for both the United States and China - the two major contributors of emissions as we saw earlier. Clearly, there is a positive correlation between CO2 and exports. But, there does seem to be a striking difference in the increase in emissions experienced by China vs that of the United States. China experienced a 20x increase

in exports and a 3x increase in emissions between 2000 and 2014. Over the same stretch, the United States experienced a 2x increase in exports, but actually an 8% **reduction** in total emissions. While China's export growth is clearly larger, one has to consider what measures taken by the US to curb emissions could have been more useful in China.




A final factor to consider here is not only the financial incentive of non-renewable energy forms, but also their current efficiency. Here we look at the relative time required to get electricity to end users (in days) encoded as size, against the total per-capita emissions of each country. One can see a rough pattern where **higher emissions countries are able to get their electricity to consumers faster (smaller, darker dots)**. Conversely, some of the lightest circles are also the biggest.

Cleaner Isn't Always Faster: Time Required to Get Electricity vs Total Emissions (2014)




This is a key factor that one must consider when trying to understand the climate change debate and logistics surrounding new policies - efficiency, speed, and price are real factors that governments consider when attempting to evoke change. Sometimes, the "faster" solution isn't always the best for the environment, which can make quick policy change difficult.

## Conclusion

Overall, this assignment gave me some great visualization practice with a real live dataset. Not only did I get better at specific Tableau tactics, but I also tried to consider different marks, encoding channels, and other visual forms. I got more exposure into how exploratory data visualization can lead to new questions, and better understanding of the data, which can further lead to additional preprocessing when necessary.

Again, here is the link to my GitHub repo fork, with pre-processing Jupyter notebook, "bad" country names, and some new filtered CSV files.